**Answer ALL questions.**

**Write your answers in the spaces provided.**

**You must write down all the stages in your working.**

**1** Write $\frac{3}{10}$ as a percentage.

30 %

**(Total for Question 1 is 1 mark)**

**2** Write the following numbers in order of size.
Start with the smallest number.

8 –7 –10 1 0 –2

–10, –7, –2, 0, 1, 8

**(Total for Question 2 is 1 mark)**

**3** Write $\frac{9}{100}$ as a decimal.

0.09

**(Total for Question 3 is 1 mark)**

**4** Write 327 correct to the nearest ten.

330

**(Total for Question 4 is 1 mark)**

**5** Write down the value of $7^2$

49

**(Total for Question 5 is 1 mark)**

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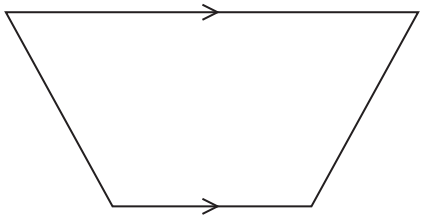
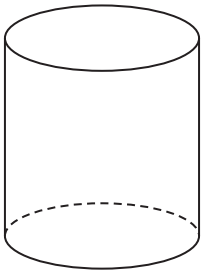
**6** (a) Write down the mathematical name of this quadrilateral.



trapezium

**(1)**

(b) Write down the mathematical name of this 3-D shape.



cylinder

**(1)**

**(Total for Question 6 is 2 marks)**

---

**7** £42 is shared equally between 3 friends.

How much does each friend get?

42 ÷ 3

$$\begin{array}{r} 14 \\ 3\overline{)42} \end{array}$$

£ 14

**(Total for Question 7 is 2 marks)**

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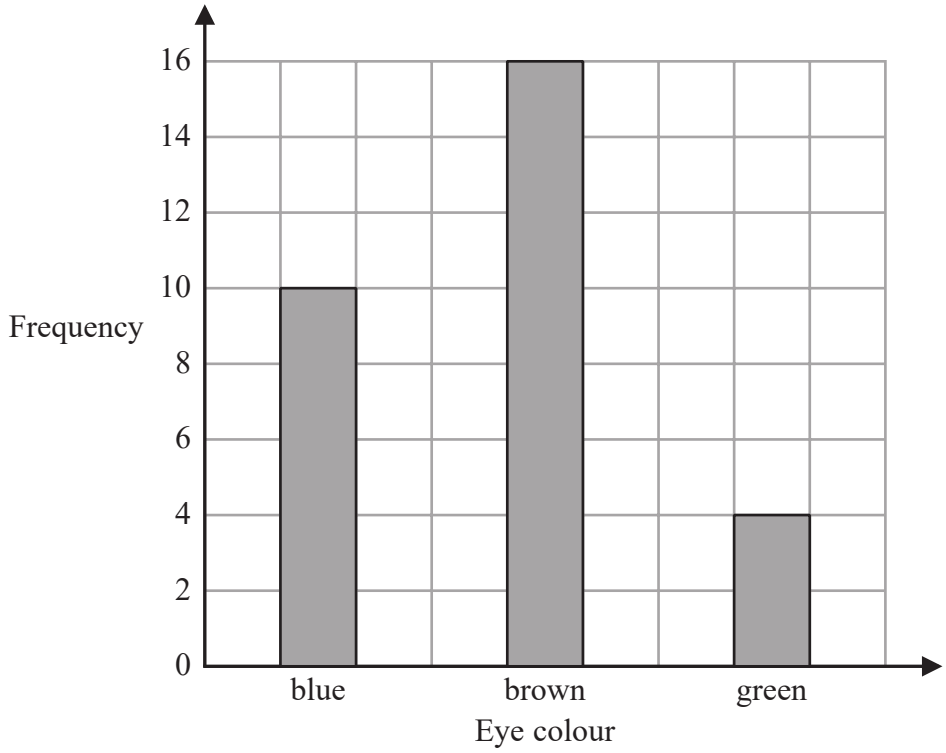
**8** Grace recorded the eye colour of each of the students in her class.

The frequency table below shows her results.

| Eye colour | Frequency |
|---|---|
| blue | 10 |
| brown | 15 |
| green | 4 |

Grace then drew the bar chart below for this information.




Write down one thing that is wrong with this bar chart.

The frequency for brown should be 15 not 16

**(Total for Question 8 is 1 mark)**

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**9** Danny buys,

1 loaf of bread for £1.20
1 bottle of milk for 70p
2 packets of cheese for £2.30 each packet

Danny pays with a £10 note.

He says,

"I should get £3.30 change."

Is Danny correct?
You must show how you get your answer.

$$\begin{array}{r} 1.20 \\ 0.70 \\ 2.30 \\ +\ 2.30 \\ \hline 6.50 \end{array}$$

$10 - 6.5 = \underline{3.5}$

£3.50 change

No.

**(Total for Question 9 is 3 marks)**

**10** Rachel records the temperature in her garden at noon each day.

On Monday, the temperature was 5 °C.
On Tuesday, the temperature was 10° less than the temperature on Monday.
On Wednesday, the temperature was 3° greater than the temperature on Tuesday.

Find the difference between the temperature on Monday and the temperature on Wednesday.
You must show all your working.

$5 - 10 = -5$

$-5 + 3 = -2$

Difference between 5°C and −2°C is 7°C

7 °C

**(Total for Question 10 is 2 marks)**



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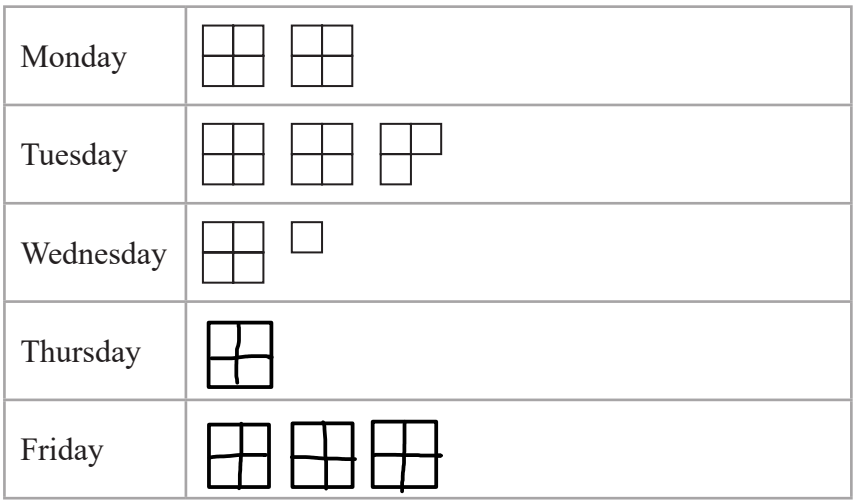
**11** The pictogram shows information about the number of video games sold in a shop on Monday, on Tuesday and on Wednesday.




**Key:**
represents 8 video games

(a) How many video games were sold on Monday?

$2 \times 8$

16

**(1)**

More video games were sold on Tuesday than on Wednesday.

(b) How many more?

Tues $11 \times 2 = 22$

Weds $5 \times 2 = 10$

$22 - 10 =$ 12

**(2)**

On Thursday and Friday, a total of 32 video games were sold in the shop.

$\frac{1}{4}$ of these 32 video games were sold in the shop on Thursday.

(c) Complete the pictogram for Thursday and Friday.

$\frac{1}{4}$ of $32 = \frac{32}{4} = 8$ (Thursday)

$32 - 8 = 24$ (Friday)

**(3)**

**(Total for Question 11 is 6 marks)**

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**12** There are two drama groups in a school.

In one group there are 36 boys and 48 girls.

In the other group, $\frac{3}{7}$ of the students are boys and the rest of the students are girls.

Ann says,

"The ratio of the number of boys to the number of girls is the same for both groups."

Is Ann correct?

You must show how you get your answer.

B : G

36 : 48

18 : 24

9 : 12

3 : 4

B : G

$\frac{3}{7} : \frac{4}{7}$

3 : 4

Yes.

**(Total for Question 12 is 3 marks)**

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**13** A number sequence starts 1 2 4

Emma says that the next term is 7

(a) Explain why Emma may be correct.

If the rule is add 1, add 2, add 3 . . .
the next term is 4+3 = 7

**(1)**

Here are the first four terms of the sequence of triangle numbers.

1 3 6 10 15 21 28 36
+2 +3 +4 +5 +6 +7 +8

(b) Find the 8th term of this sequence.

36

**(2)**

**(Total for Question 13 is 3 marks)**

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P 6 4 6 2 9 A 0 8 2 4

**14** 3 kg of carrots cost £1.80
2 kg of carrots and 5 kg of potatoes cost a total of £3.45

Work out the total cost of 4 kg of carrots and 2 kg of potatoes.
You must show all your working.

3 kg of c £1.80 ÷3
1 kg of c £0.60 ×2
2 kg of c £1.20
4 kg of c £2.40

3.45 − 1.20
= 2.25

5 kg of p £2.25

$5\overline{)2^{2}2^{2}5}$ = 0.45

1 kg of p £0.45
2 kg of p £0.90

2.40
+ 0.90
3.30

£ 3.30

**(Total for Question 14 is 4 marks)**



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**15** (a) Expand $2(a + d)$

$2a + 2d$

**(1)**

(b) Factorise $6y^2 - 5y$

$y(6y - 5)$

**(1)**

(c) Solve $4x - 7 = 37$

$+7 \quad +7$

$4x = 44$

$\div 4 \quad \div 4$

$x = 11$

$x =$ 11

**(2)**

**(Total for Question 15 is 4 marks)**

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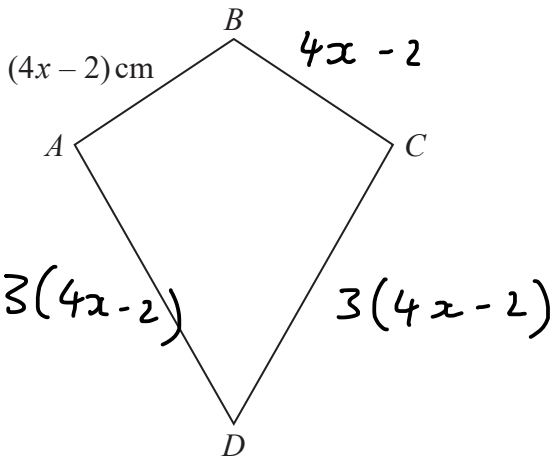
**16** *ABCD* is a kite.




$AB = (4x - 2)$ cm

Jasper says that $x$ could be 0.5

(a) Explain why Jasper cannot be correct.

$4(0.5) - 2 = 0$

The length cannot be zero.

**(1)**

$AD = 3AB$

The kite has a perimeter of 64 cm.

(b) Find the value of $x$.

$8(4x-2) = 64$

$\div 8 \qquad \div 8$

$4x - 2 = 8$

$4x = 10$

$x = \frac{10}{4} = \frac{5}{2} = 2.5$

$x =$ 2.5

**(3)**

**(Total for Question 16 is 4 marks)**



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**17** Heidi wants to make some biscuits using this recipe.

| Makes 12 biscuits |
|---|
| 125 g butter |
| 200 g flour |
| 50 g sugar |

Heidi thinks that she has,

500 g butter
700 g flour
250 g sugar

Assuming that these weights are correct,

(a) work out the greatest number of biscuits Heidi can make.
You must show all your working.

Butter $\frac{500}{125} = 4$ $\quad 12 \times 4 = 48$

Flour $\frac{700}{200} = \frac{7}{2}$ $\quad 12 \times \frac{7}{2} = 6 \times 7 = \underline{\underline{42}}$

Sugar $\frac{250}{50} = 5$ $\quad 12 \times 5 = 60$

42

**(4)**

Heidi is wrong.
She has more than 250 g of sugar.

(b) Does this affect the greatest number of biscuits Heidi can make?
Give a reason for your answer.

No. Heidi only has enough Flour for 42 biscuits, more sugar would not change this.

**(1)**

**(Total for Question 17 is 5 marks)**



P 6 4 6 2 9 A 0 1 2 2 4

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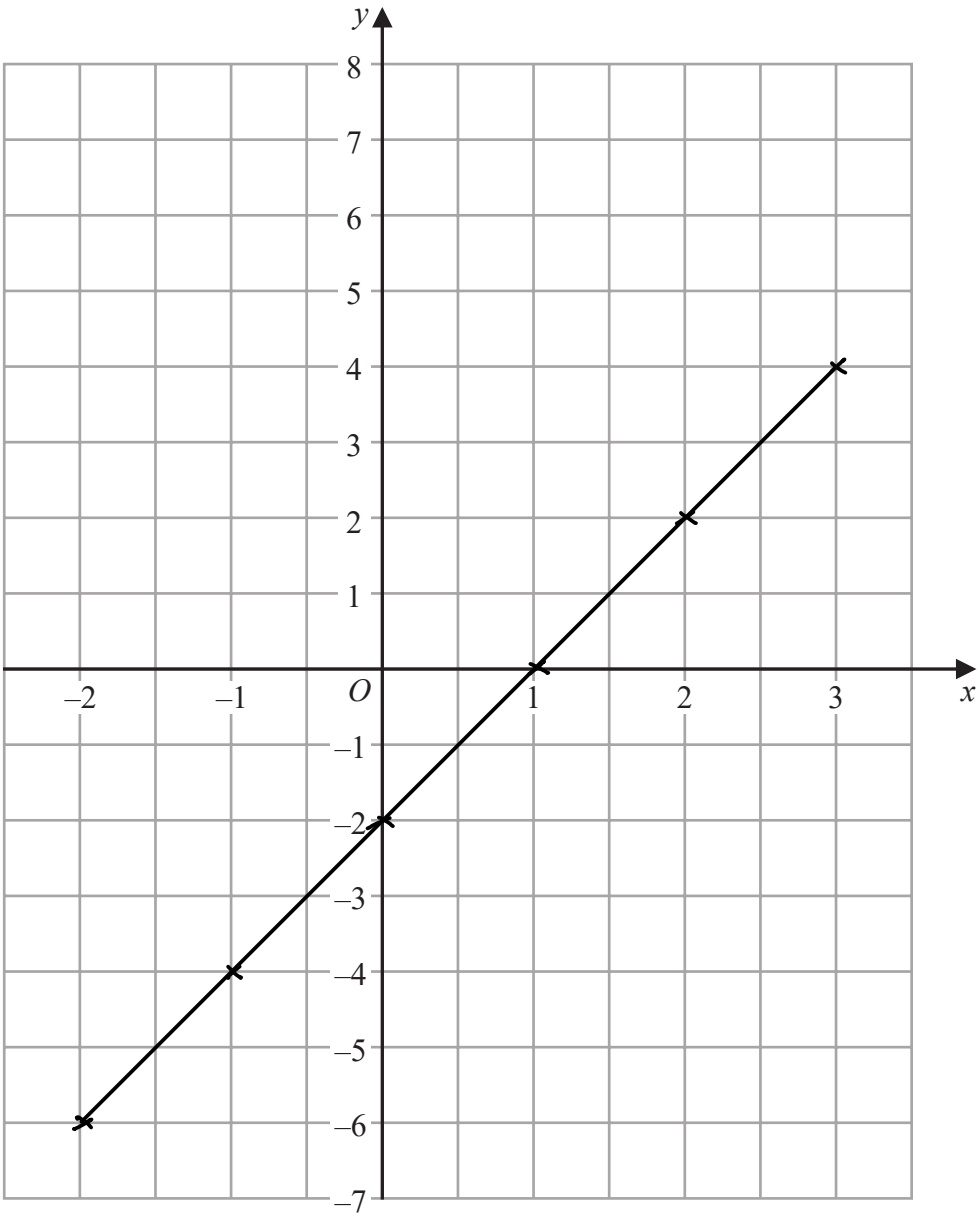
**18** On the grid below, draw the graph of $y = 2x - 2$ for values of $x$ from –2 to 3

| $x$ | -2 | -1 | 0 | 1 | 2 | 3 |
|---|---|---|---|---|---|---|
| $y$ | -6 | -4 | -2 | 0 | 2 | 4 |




**(Total for Question 18 is 3 marks)**



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**19** Robin buys a watch for £80
He sells the watch for £56

Work out his percentage loss.

$\frac{\text{change}}{\text{original}} \times 100$

$80 - 56 = 24$

$\frac{24}{80} \times 100$

$\boxed{\frac{24}{80} \begin{matrix} \div 8 \\ \div 8 \end{matrix} = \frac{3}{10}}$

$\frac{3}{10} \times 100 = 30$

30 %

**(Total for Question 19 is 3 marks)**

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**20** (a) Work out $3.67 \times 4.2$

$$\begin{array}{r} 367 \\ \times\ 42 \\ \hline 734 \\ +\ 1468\phantom{0} \\ \hline 15414 \end{array}$$

15.414

**(3)**

(b) Work out $59.84 \div 1.6$

$$\frac{59.84}{1.6} = \frac{598.4}{16}$$

16
32
48
64
80
96
112

$$16 \overline{)\,5\,{}^{5}9\,{}^{11}8\,.\,{}^{6}4} \quad \text{037.4}$$

37.4

**(3)**

**(Total for Question 20 is 6 marks)**



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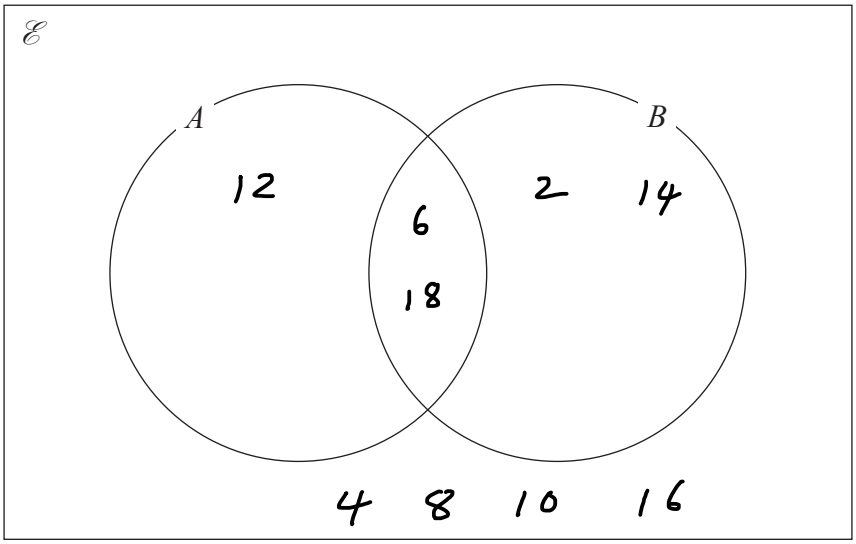
**21** $\mathscr{E}$ = {even numbers less than 19}
$A$ = {6, 12, 18}
$B$ = {2, 6, 14, 18}

Complete the Venn diagram for this information.




**(Total for Question 21 is 3 marks)**

**22** Work out $4\frac{1}{5} - 2\frac{2}{3}$

Give your answer as a mixed number.

$4 = \frac{20}{5}$ $2 = \frac{6}{3}$

$\frac{21}{5} \times 3 - \frac{8}{3} \times 5$

$\frac{63}{15} - \frac{40}{15} = \frac{23}{15} = 1\frac{8}{15}$

$1\frac{8}{15}$

**(Total for Question 22 is 3 marks)**

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**23** At the end of 2017

the value of Tamara's house was £220 000
the value of Rahim's house was £160 000

At the end of 2019

the value of Tamara's house had decreased by 20%
the value of Rahim's house had increased by 30%

At the end of 2019, whose house had the greater value?
You must show how you get your answer.

Tamara

$10\% = \frac{220000}{10} = 22000$

$20\% = 2 \times 22000$
$= 44000$

$$\begin{array}{r} 220000 \\ -\ 44000 \\ \hline 176000 \end{array}$$

Rahim

$10\% = \frac{160000}{10} = 16000$

$30\% = 3 \times 16000$
$= 48000$

$$\begin{array}{r} 160000 \\ +\ 48000 \\ \hline 208000 \end{array}$$

Rahim's House

**(Total for Question 23 is 4 marks)**



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**24** Rosie, Matilda and Ibrahim collect stickers.

$$\frac{\text{number of stickers}}{\text{Rosie has}} : \frac{\text{number of stickers}}{\text{Matilda has}} : \frac{\text{number of stickers}}{\text{Ibrahim has}} = 4:7:15$$

Ibrahim has 24 more stickers than Matilda.

Ibrahim has more stickers than Rosie.
How many more?

15 − 7 = 8

8 parts = 24 stickers

1 part = 3 stickers

R : M : I

4 : 7 : 15

×3 ×3 ×3

12 : 21 : 45

45 − 12 = 33

33

**(Total for Question 24 is 3 marks)**

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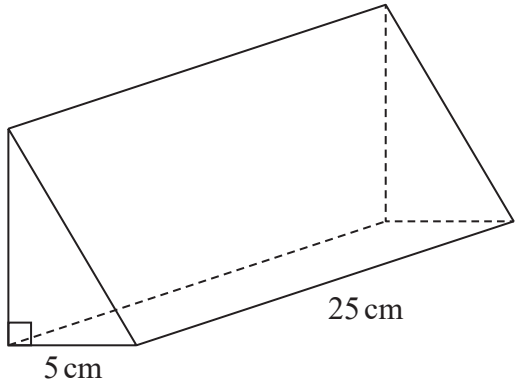
**25** The diagram shows a prism.




The cross section of the prism is a right-angled triangle.
The base of the triangle has length 5 cm

The prism has length 25 cm
The prism has volume 750 $cm^3$

Work out the height of the prism.

$\frac{1}{2} \times \text{base} \times \text{height} \times \text{length} = \text{volume}$

$\frac{1}{2} \times 5 \times h \times 25 = 750$

$\frac{125}{2} h = 750$

$125h = 1500$

$h = \frac{1500}{125}$

$= 12$

12 cm

**(Total for Question 25 is 3 marks)**



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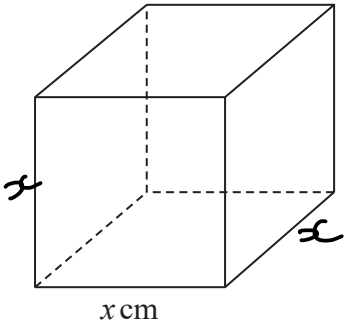
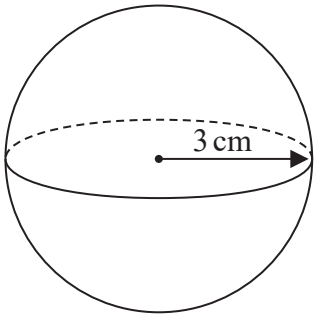
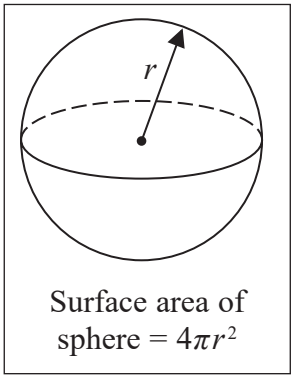
**26** The diagram shows a cube with edges of length $x$ cm and a sphere of radius 3 cm.




The surface area of the cube is equal to the surface area of the sphere.

Show that $x = \sqrt{k\pi}$ where $k$ is an integer.

Surface Area of Cube $= 6x^2$

Surface Area of Sphere $= 4\pi(3)^2$

$= 4\pi(9)$

$= 36\pi$

$6x^2 = 36\pi$

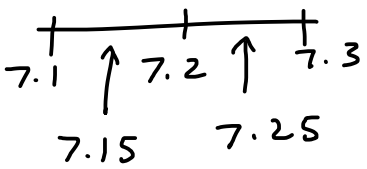
$x^2 = 6\pi$

$x = \sqrt{6\pi}$

**(Total for Question 26 is 4 marks)**

**27** Freddie measured the length of a pencil as 7.2 cm correct to 1 decimal place.

Complete the error interval for the length, $p$ cm, of the pencil.




7.15 $\leqslant p <$ 7.25

**(Total for Question 27 is 2 marks)**



P 6 4 6 2 9 A 0 2 0 2 4

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**28** The equation of a straight line **L** is $y = 3 - 4x$

(i) Write down the gradient of **L**.

$y = mx + c$

$y = -4x + 3$

$m$ $c$

$-4$

**(1)**

(ii) Write down the coordinates of the point where **L** crosses the $y$-axis.

( 0 , 3 )

**(1)**

**(Total for Question 28 is 2 marks)**

**TOTAL FOR PAPER IS 80 MARKS**

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